

Exercise 8: Reinforcement Learning

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Problem 1 (Discounted Sums Sanity Check):

A reward stream satisfies $r_t = 1$ for all $t \geq 0$. For $\gamma \in [0, 1)$:

1. Compute $G_0 = \sum_{k=0}^{\infty} \gamma^k$.
2. Compute $G_5 = \sum_{k=0}^{\infty} \gamma^k$ (the return starting at time $t = 5$).
3. State what happens to G_0 as $\gamma \rightarrow 1^-$.

Problem 2 (Evaluating a Fixed Policy in a 2-State MDP):

Two states s_1, s_2 ; actions A, B in both. Deterministic dynamics/rewards:

$$s_1: A \rightarrow s_1, r = 0; \quad B \rightarrow s_2, r = 2.$$

$$s_2: A \rightarrow s_1, r = 1; \quad B \rightarrow s_2, r = 0.$$

Let $\gamma = 0.5$ and policy parameters $\pi(A|s_1) = \alpha$, $\pi(B|s_1) = 1 - \alpha$, $\pi(A|s_2) = \beta$, $\pi(B|s_2) = 1 - \beta$. Write Bellman expectation equations for $v_\pi(s_1), v_\pi(s_2)$ and solve in closed form in terms of α, β .

Problem 3 (Contraction of the Optimality Operator):

Define $(Tv)(s) = \max_a \sum_{r,s'} p(r, s'|s, a) [r + \gamma v(s')]$. Prove $\|Tv - Tw\|_\infty \leq \gamma \|v - w\|_\infty$.

Problem 4 (Baseline Unbiasedness):

In policy gradient methods, one often subtracts a baseline $b(s_t)$ from the return G_t to reduce the variance of the gradient estimate:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a_t|s_t) (G_t - b(s_t))].$$

Show that this modification does *not* introduce any bias in the estimator, provided that $b(s_t)$ does not depend on the sampled action a_t . That is, prove that:

$$\mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(a_t|s_t) b(s_t)] = 0.$$

Then briefly explain why subtracting such a baseline can still be beneficial in practice.